

MODULE 06 · DOMAIN 2 - DATA INGESTION AND LOADING

Lakeflow Connect

Managed and standard connectors, the three-layer stack, ingestion pipelines and gateways, and the rule for choosing a layer.

DOMAIN 2 of 7 EXAM WEIGHT 21% MODULE 06 of 36

Lakeflow Connect is not one tool. It is an **umbrella over the whole ingestion stack**, and questions about it are almost always asking which layer of that stack you should use.

The connector types

TYPE	WHAT IT INGESTS FROM
Database (CDC)	Relational databases using change data capture: MySQL, PostgreSQL, SQL Server
Query-based	Databases, by querying the source directly, with no CDC configuration
SaaS	Enterprise applications: Salesforce, Workday, HubSpot, Jira and others
File source	Enterprise file services such as SharePoint and Google Drive
Streaming	Message buses and event streaming sources
Community and custom	Sources with no built-in connector

ON THE EXAM

Note that database sources have two routes: CDC-based and query-based. They have different architectures and different costs.

The three layers of the ingestion stack

01

Structured Streaming

Most customisable. Full control, full Spark APIs, you build everything.

02

Lakeflow pipelines

Declarative. Orchestration, monitoring, quality and error handling are managed.

03

Managed connectors

Most automated. Auth, CDC, retries and schema evolution all handled.

LEARN THIS RULE

Databricks' rule: start with the most managed layer. If it does not satisfy your requirements, drop down to the next.

Why the rule runs downward, not upward

- ◆ **Each layer is built on the one below it.** Managed connectors run on pipelines, which run on Structured Streaming.
- ◆ **So moving down never adds capability you lose by moving up.** It adds work you would otherwise not do.
- ◆ **Starting at the bottom means rebuilding solved problems:** auth, retries, schema drift, incremental state.
- ◆ **The only good reason to drop down is a requirement the layer above cannot meet.**

ON THE EXAM

In an exam, the most managed option that meets every stated requirement is usually correct. Extra control you were not asked for is a cost.

Managed or standard connectors?

MANAGED CONNECTORS

Fully handled

- Source-specific auth, CDC and edge cases handled
- Governed by Unity Catalog, on serverless compute
- Automatic retries with exponential backoff
- Limited to the sources Databricks has built

STANDARD CONNECTORS

More control, more work

- Broader source support
- Choose your own level: Structured Streaming or pipelines
- You configure discovery, schema handling and state
- Auto Loader is the canonical standard connector

THE TIEBREAKER

Standard connectors trade automation for broader source support and customisation. That trade is the whole distinction.

Anatomy of a managed connector

COMPONENT	WHAT IT IS
Connection	A Unity Catalog securable holding the source's authentication details
Ingestion pipeline	Reads the source and writes the results. Runs on serverless compute by default.
Destination tables	Streaming tables that the pipeline writes into
Ingestion gateway	Database CDC connectors only. Captures changes continuously.
Staging storage	Database CDC connectors only. Holds captured changes.

ON THE EXAM

SaaS, file, query-based and streaming connectors use the first three. Only database CDC connectors add a gateway and staging.

Two ways to ingest from a database

DATABASE CONNECTOR

Change data capture

- An ingestion gateway runs continuously as its own job
- Staging storage holds the captured changes
- Captures every change, including deletes
- The gateway is continuous, so not fully serverless

QUERY-BASED CONNECTOR

Query the source on a schedule

- No gateway, no staging storage
- The pipeline queries the source directly
- Simpler and cheaper to run
- Sees the state at query time, not every change between runs

THE TIEBREAKER

CDC catches everything that happened. Query-based catches what is there when it looks. That difference decides which you need.

How incremental ingestion works here

- ◆ **The first run ingests everything selected** from the source, and starts tracking changes in parallel.
- ◆ **Every later run uses that change tracking** to ingest only what changed since the last run.
- ◆ **The approach depends on the source.** SQL Server can use both change tracking and CDC; Salesforce picks a cursor column.
- ◆ **Some sources and tables do not support incremental ingestion yet.** Those fall back to a full load each run.

ON THE EXAM

Without a valid cursor or change-tracking mechanism, the connector performs a full load every run. That is the cause behind a whole family of cost complaints.

Orchestration

- ◆ Add a schedule to a pipeline and Lakeflow Connect creates a job for it automatically.
- ◆ The ingestion pipeline becomes a task inside that job, and you can add further tasks around it.
- ◆ A pipeline can have more than one schedule, each getting its own job.
- ◆ For database connectors the gateway runs in its own separate job, as a continuous task.

ON THE EXAM

That last point explains the cost profile. The pipeline may run four times a day, but the gateway never stops.

Failure recovery

- ◆ **Failures retry automatically with exponential backoff.** Most transient problems resolve without anyone noticing.
- ◆ **Some errors need you,** most commonly expired credentials.
- ◆ **The connector stores the last cursor position,** so it resumes from there rather than losing data.
- ◆ **A full refresh can erase connector-generated history.** Treat it as destructive, not as a reset button.

ON THE EXAM

Automatic recovery is a large part of what managed connectors are for. It is also why 'we can just write a notebook' underestimates the work.

Networking

- ◆ **SaaS connectors reach out to the source's APIs**, and are automatically compatible with serverless egress controls.
- ◆ **Cloud databases** can connect over Private Link, or the gateway can be deployed inside a peered VNet or VPC.
- ◆ **On-premises databases** connect through services such as AWS Direct Connect or Azure ExpressRoute.
- ◆ **Networking is usually the real constraint** on whether a managed connector is viable, rather than the connector itself.

ON THE EXAM

When a scenario mentions an on-premises source behind a corporate firewall, the network path is the thing to check first.

Standard connectors, by level of control

SOURCE	MORE CUSTOMISATION	MORE AUTOMATION
Cloud object storage	Auto Loader with Structured Streaming	Auto Loader with Databricks SQL
Apache Kafka	Structured Streaming with Kafka source	Databricks SQL with Kafka source
Google Pub/Sub	Structured Streaming with Pub/Sub source	Lakeflow pipelines with Pub/Sub source
SFTP servers	Python or SQL	Not available

ON THE EXAM

The same source appears at several levels. Choosing a standard connector means choosing a source and a level of control.

The SQL route

- ◆ **CREATE STREAMING TABLE** is the SQL syntax for incremental ingestion from cloud object storage.
- ◆ Databricks calls it the **recommended alternative to COPY INTO** for SQL users.
- ◆ It **wraps Auto Loader**, so all the schema and discovery behaviour from module 5 still applies.
- ◆ **Schema location and checkpoints are managed for you** when it runs inside a Lakeflow pipeline.

ON THE EXAM

This is the same recommendation module 04 flagged, seen from the other side. It does not make COPY INTO a wrong answer.

Deployment and connections

- ◆ **Ingestion pipelines deploy with Declarative Automation Bundles**, so they get source control, review and CI/CD.
- ◆ **Connections can be created programmatically** via the Connections API, the CLI, or a bundle pre-deploy script.
- ◆ **Except where browser-based OAuth is the only option.** Those need an interactive sign-in and cannot be automated.
- ◆ **Affected connectors include** Jira, HubSpot, Slack, Zendesk Support and several advertising platforms.

ON THE EXAM

A fully automated deployment can still be blocked by one connector that requires a human to click through a sign-in.

Two risks the documentation states plainly

- ◆ **Managed connectors depend on external services** that Databricks does not control.
- ◆ **Databricks may discontinue a connector** if changes to that external service make it impractical to maintain.
- ◆ **Connectors sit in various release states.** Check whether one is generally available before designing around it.
- ◆ **Monitoring is provided:** event logs, cluster logs, pipeline health and data quality metrics, plus `system.billing.usage` for cost.

ON THE EXAM

For database connectors specifically, gateway progress can be monitored in real time through event logs.

Lakeflow Connect or Lakehouse Federation?

LAKEFLOW CONNECT

Ingest a copy

- Reads the source and writes data into Delta tables
- The data now lives in your lakehouse
- Queries do not touch the source system
- Freshness depends on the pipeline schedule

LAKEHOUSE FEDERATION

Query in place

- Registers the source as a foreign catalog
- No copy is made; data stays where it is
- Every query reaches out to the source system
- Always current, but load lands on the source

WORTH DISTINGUISHING

Federation is not a Domain 2 objective on the current exam, but the distinction is a clean confusion pair worth holding.

Tiebreakers from this module

WHEN A QUESTION SAYS	REACH FOR
Salesforce, Workday, ServiceNow or similar	A managed SaaS connector
Every change to the database, including deletes	A database CDC connector, with a gateway
Periodic snapshot of database tables, cheaply	A query-based connector, no gateway
Files in cloud object storage	Auto Loader, a standard connector
Kafka, Kinesis or a message bus	Structured Streaming or a streaming connector
A source with no built-in connector	Community, then custom
The source must not be copied	Lakehouse Federation, not ingestion

ON THE EXAM

Seven lines. Most Lakeflow Connect questions are answered by naming the source type correctly.

A marketing team needs Salesforce opportunity and account data in the lakehouse, refreshed several times a day. Nobody on the team writes Python. The platform team requires that credentials are governed and that the destination tables appear in Unity Catalog with lineage.

An engineer offers to write a notebook that calls the Salesforce REST API and handles pagination, retries and token refresh himself.

How should the team proceed?

- ☐ A Accept the notebook, since it gives full control over the API calls
- ☐ B Use the managed Salesforce connector, which handles auth, incremental reads and retries
- ☐ C Use Auto Loader pointed at a directory the notebook exports to
- ☐ D Use Lakehouse Federation to query Salesforce in place

◆ PAUSE AND DECIDE

How should the team proceed?

A Accept the notebook

It rebuilds auth, pagination, retries and incremental state by hand, and nobody on the team can maintain it. Control nobody asked for.

C Auto Loader over an exported directory

Keeps the fragile notebook and adds a second hop. Auto Loader reads files, so something still has to produce them.

B The managed Salesforce connector

Correct. A managed connector exists for exactly this source. Credentials live in a UC connection, and destination tables are governed streaming tables.

D Lakehouse Federation

Queries in place rather than ingesting, so every downstream query hits Salesforce. The team asked for data in the lakehouse.

THE SIGNAL

Start at the most managed layer. A managed connector exists for this source and meets every stated requirement, so nothing justifies dropping down.

A finance team ingests from an on-premises SQL Server using a managed database connector. The ingestion pipeline is scheduled four times a day. Reviewing costs, an analyst is puzzled: the pipeline runs for about twelve minutes each time, but compute charges look continuous.

She asks whether the pipeline has been misconfigured to run continuously.

What should she be told?

- A The pipeline is misconfigured; it should be changed from continuous to triggered mode
- B The ingestion gateway runs continuously in its own job to capture changes, separate from the scheduled pipeline
- C Serverless compute bills continuously once enabled, regardless of pipeline activity
- D The destination streaming tables keep a cluster alive to serve queries

◆ PAUSE AND DECIDE

What should she be told?

- A

 Change the pipeline to triggered mode
The pipeline is already triggered, four times a day. Changing a setting that is already correct will not touch the continuous cost.
- B

 The gateway runs continuously in its own job
Correct. Database CDC connectors deploy a gateway as a continuous task in a separate job, because change capture cannot be intermittent.
- C

 Serverless bills continuously once enabled
False, and the reverse of how serverless works. It bills for what runs, which is why it suits bursty workloads.
- D

 Streaming tables keep a cluster alive
A streaming table is a table. It does not hold compute open to serve queries.

THE SIGNAL

The cost is not a fault. Continuous change capture is the architecture, and it is the price of catching every change including deletes.

A team needs to ingest events from a self-hosted Apache Kafka cluster into bronze. They need custom deserialisation of a proprietary binary payload, and they must apply a per-record routing rule before anything is written.

A colleague argues that Databricks recommends starting with the most managed layer, so they should look for a managed connector first.

What is the right conclusion?

- ☐ A The colleague is right; the rule always means choosing the most managed option available
- ☐ B The rule applies, but custom deserialisation and per-record routing are requirements the managed layer cannot meet, so drop down
- ☐ C The rule does not apply to message buses, which always require Structured Streaming
- ☐ D Kafka cannot be ingested by Databricks; the data must first be written to object storage

◆ PAUSE AND DECIDE

What is the right conclusion?

- A

 Always choose the most managed option
Drops the second half of the rule. Databricks says start at the most managed layer and drop down when it does not satisfy your requirements.
- B

 The rule applies, but these requirements force a drop down
Correct. Custom deserialisation and per-record routing need code, which is exactly the situation the rule's second clause describes.
- C

 The rule does not apply to message buses
It does. Kafka appears at several levels, including Lakeflow pipelines and Databricks SQL. The rule is not suspended by source type.
- D

 Kafka cannot be ingested directly
False. Kafka is a supported source at multiple levels of the stack.

THE SIGNAL

The rule has two clauses. Candidates who memorise only the first one get this wrong in exactly the way the colleague does.

A managed connector ingests eleven tables from a SaaS application. Ten of them refresh in under a minute each run. The eleventh takes forty minutes every time and its row count in the destination is identical after every run.

The pipeline is configured identically for all eleven tables.

What is the most likely explanation?

- ☐ A That table is much larger, so it simply takes longer
- ☐ B That table has no valid cursor or change-tracking mechanism, so the connector performs a full load every run
- ☐ C The ingestion pipeline needs a separate schedule for that table
- ☐ D The destination table needs OPTIMIZE to speed up the writes

◆ PAUSE AND DECIDE

What is the most likely explanation?

- A

 It is simply a larger table
Size alone would make the first run slow. Incremental runs afterwards should be proportional to what changed, not to total size.
- C

 It needs its own schedule
Scheduling changes when the work happens, not how much work there is. Forty minutes would still be forty minutes.
- B

 No valid cursor, so a full load every run
Correct. Some sources and tables do not support incremental ingestion. Without a cursor, the connector reloads everything each time.
- D

 Run OPTIMIZE on the destination
File layout affects reads. It does not explain why this one table reprocesses its whole contents on every run.

THE SIGNAL

'Identical row count after every run' is the clue. Consistent full reloads look identical; genuine incremental loads do not.

Start at the most managed layer. Drop down only when a stated requirement forces it. **Both halves of that sentence are examinable.**